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Chapter 3

Configurator Overview



Introduction

The Configurator Overview explains the following:

- What Configurator is.
- How Configurator fits with other modules.
- The benefits that Configurator provides with regard to order entry.
- The benefits with regard to production control.
- How Configurator works
 - these paragraphs are based on a simple configuration which is used as an example throughout the product information.

What is Configurator?

If your company tailors a product to the exact requirements of a customer, and assembles or manufactures the product to conform to a specific customer order, then you need a system which presents choices and controls permutations while the order is being placed. The system should then use the information to create the production order.

Configurator displays all the different options available for the product when your customer calls to place an order. You can set up rules to ensure that incompatible orders are not created - for example, if your customer selects a PVC desk this cannot also have the varnish finish which would be available for the pine desk.

Once configured, the price of the product is confirmed, allowing product margin control. The system automatically creates a confirmed production order, which shows all the components and operations required.

Configurator means that you can set up a single production route for your basic product, and then specify all available variations (materials, colour etc.) within that same route.

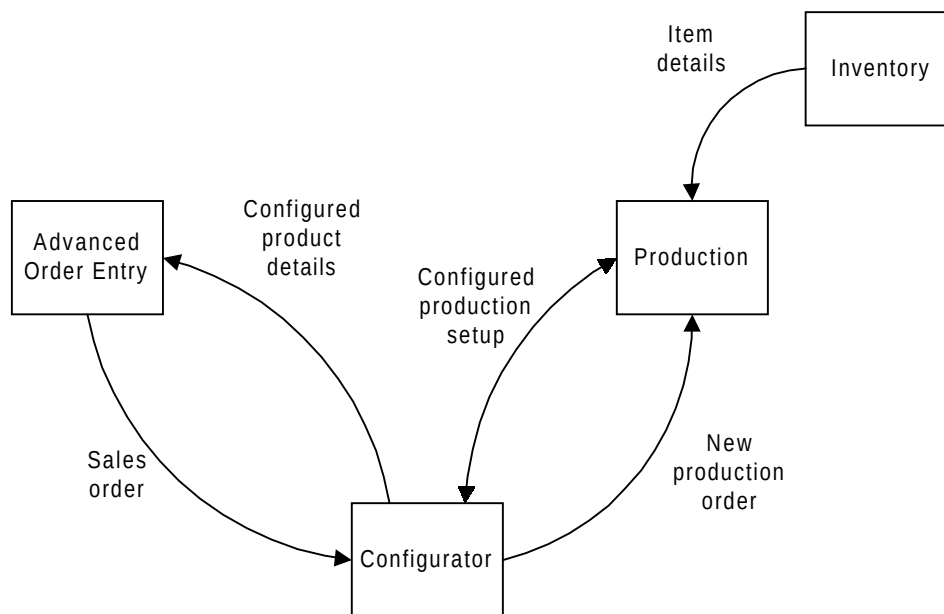
Accurate configuration leads to high customer satisfaction, but the route needs careful control if an accurate configuration is to be created first time and pass without error into production. It is in these areas that Configurator can provide substantial benefits.

Configurator and Other Applications

The main functions of Configurator are:

- To enable a sales order to be easily configured according to options displayed when the order is being placed.
- To create a production order to fulfil the sales order.
- To calculate costs and prices according to the costing and pricing methods selected.
- To facilitate the setting up and maintenance of items and routes.

The basic interfaces are illustrated below:



Prerequisites

You must have the following Distribution and Production modules:

- Inventory Management
- Sales Order Processing
- Advanced Order Entry
- Production Definition Management
- Production Control

Other Related Applications

If you use the following modules with Configurator, it provides a comprehensive product set for the configure-to-order marketplace.

- Customer Returns - allows configured products to be received and optionally matched against the original invoice to enable accurate costs and credit notes to be generated.
- Sales Analysis - enables invoice details to be reviewed for fully configured products.
- Transport Planning - plans the physical storage and distribution of configured products by assuming that the fluctuation in pack sizes, weights and volume between variants of the configured product is negligible.
- Warehouse Management - plans the physical storage and distribution of configured products by assuming that the fluctuation in pack sizes, weights and volume between variants of the configured product are negligible.
- Advanced Financial Integrator (AFI) - you use this to update user-defined General Ledger accounts with accurate stock and cost of sale values.

The Benefits in Order Entry

- All the configuration options are presented clearly and in order.
- All the information necessary for creating the sales order (and a production order, if required) is collected first time. This reduces the likelihood of error, and provides customers with a more efficient service.
- Product permutations are easily controlled. For example, if a customer orders a pine desk a choice of paint colours is available, but if the customer selects a PVC desk, a paint finish is not offered.
- The presentation of options can be controlled by user ID, so that it suits the experience and knowledge of each operator.
- Enquiries provide easy tracking of the progress of your sales and production orders.
- Three methods of cost calculation are offered:
 - A price list price for the model is used unchanged for all items configured from that model.
 - The price is the sum of the price list prices for all components.
 - The price is the sum of the cost list prices for all components, plus operation costs (optional), plus a percentage mark-up as required.

The Benefits in Production Control

When setting up the route

- If you are using the preconfiguration process, you need only set up and maintain one model item and route, instead of items and routes for individual items.
- Once your generic item (e.g. 'Paint') and your actual items (e.g. 'green paint', 'blue paint' etc.) are set up, you need maintain only the generic item.

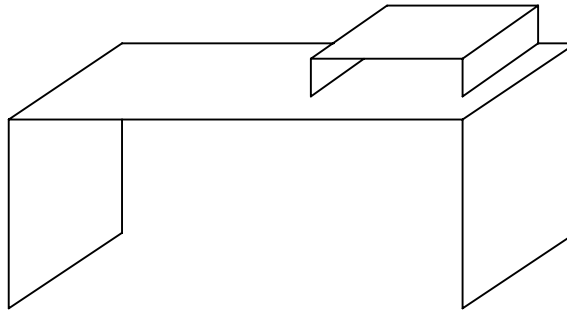
When creating the sales order

- Once the sales order is created, a production order is created automatically to fulfil the order.
- Configurator rules govern the creation of the sales and production order, so that invalid orders are not created. The rules can ensure, for example, that if a desk has a green top, the sides are also green.

How Configurator Works

The working of the Configurator depends to a great extent on the careful and accurate organisation of items and routes within the Inventory and Production applications. This in itself depends on a good understanding of how the information will be used and presented by Configurator.

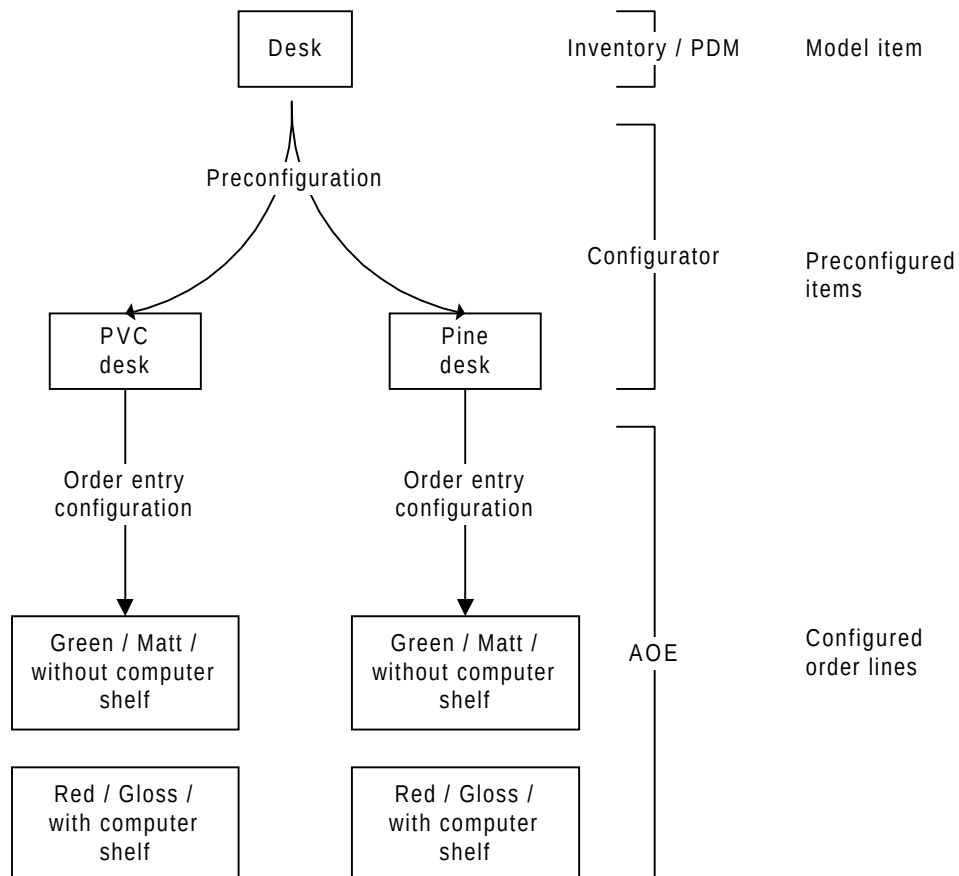
The following paragraphs outline the working methods of Configurator, and introduce essential vocabulary. The concepts are illustrated by a simple example, a child's desk. This is illustrated below:



Once you understand the ideas and terms used, you are in a position to turn to the Implementation procedure, which takes you through the process step by step.

What is Preconfiguration?

You can select some options before the sales order is placed. If, for example, you have set up the desk as the model item but wish to have the pine desk and the PVC desk separated as items before the customer places the order, you can preconfigure. The diagram below shows the position of preconfiguration in the application:



On completion of preconfiguration the following item master records are automatically created for the preconfigured product:-

- Inventory item
- Inventory item/stockroom
- Production item

The item code itself can be generated automatically or manually.

What is Order Configuration?

If you do not wish to use the preconfiguration function then you are saved the work of setting up a model and base items. All you have to do is to set up your item as an order configured item in Inventory, and attach a route to it (this route must be the configuration route as defined in the Configurator company profile).

All the options and features you then define for the item are presented for selection during order entry:

Options are presented in selection windows, but the display can be tailored to an individual operator, depending on their level of experience and expertise in configuring products.

You can also reselect any number of components during discussion or negotiation with the customer, as the price of the particular configuration is displayed on a review screen.

If required, Configurator displays all optional operations and features not yet added to the configuration. You can select any of these to add to the configuration, and then return to the configuration review screen and complete the configuration.

Defining the Item

Within standard Production, a parent item, defined within Inventory and Production Definition Management (PDM), is attached to a route which specifies all the operations and components necessary to produce that item.

A parent item which you want to configure is defined in Inventory and PDM in just the same way, except that the setting of a flag in Inventory defines the item as a model item. This is the unconfigured basic item whose route holds all the definitions for any configured variation. In our example it is the desk itself.

The model item is attached to the route which holds its operations and components. This is, in every case, the configuration route as specified in the Configurator company profile. This is the only route which can be used by Configurator for the company in question.

If the configuration route for the model item contains many features and rules which you would like to use for other items, you can set up base items. Base items use the same route, including operations and components, as the model item with which they are associated, but because they are separate Inventory items their Inventory set up can differ from that of the model item.

Defining the Route

The normal series of operations and components is entered, but the route must hold extra information. Both components and operations can offer options.

Below is the route for our desk, with the configurations available on this route added as comments:

OPERATION/ COMPONENT		OPTIONS
0010	Assemble desk	
0010	1 x Left Side	The sides and top must be offered in the full range of colours and finishes.
0020	1 x Right Side	
	1 x Top	The colour and finish of both sides and top must match.
		The top can be plain, or it can have a cable hole for the fixing of a computer shelf (optional)
0030	4 x Screws	
0020	Paint desk	This operation is optional, depending on colour chosen.
0010	Paint	Must match desk colour chosen.
0030	Fix computer shelf	
0010	Computer shelf in black PVC	If the computer shelf is not ordered, this operation is not performed.

Characteristics

Our desk is available in a choice of materials (pine and PVC), and a choice of paint finish (matt, gloss) and paint colour (blue, green, and red paint, and clear varnish).

In order to control the options that can be selected, this information must be set up carefully.

The following diagram shows how the three 'areas' of choice (material, finish and paint colour) are entered on the system:

The options available here are to do with the choice of paint, which is set up as a search family. Within this, the choices of finish and colour are called characteristics, to be set up via the Inventory Descriptions option under a major type.

'Paint' itself is set up as an item in Inventory, of the type to be known as a generic item, with its own item code. Each item within this generic item must also be set up in Inventory in the same way as any other item. These items must include every valid permutation of characteristics:

PAINT	Matt	Gloss
Red	Y	Y
Blue	Y	Y
Green	Y	Y
Clear	N	Y

You can see from the above table that seven permutations must be allowed. These, once set up, are known as actual items.

Rules and Messages

You notice that the selection of different characteristics in itself involves complications. If you select a computer shelf then the shelf fix operation is required, while if you do not, then the operation is not required. All such requirements and any invalid permutations are specified in rules, and if a rule is broken a message can be displayed.

Rules are set up and maintained separately from the route itself, but are 'invoked' within the route, as are messages.

Creating the Production Order

On completing the sales order, a confirmed production order is created for each configured sales order line, containing all selected operations and optional components as well as any standard components.

Any internal sales order line text, plus any text entered during configuration is also transferred on to the production order.

Whilst the production order is held, i.e. not yet released, the configuration for the product can be amended by reconfiguring the product in Advanced Order Amend.

A default planning filter is added automatically to the production order, so you can prevent MPS and/or MRP from replanning the configured product.



Note: If Configurator is installed, then back-to-back production orders may also be created for 'standard products' using the default route for the product, i.e. without any configuration involved.

Allocate Finished Product

Once the production order is released for the configured product, then Production Control handles it in almost the same manner as a normal production order, with the following differences:

If you try to amend any key details on the production order, in particular quantity, date or stockroom, then a message tells you that the production order is linked to a sales order line. An exception record is also created.

If the production order is to be released or booking in of finished goods is attempted whilst the associated sales order is suspended, cancelled or a quotation, then again a message is issued (and an exception record written).

Full visibility of linked sales order lines is provided via windows displayed within these transactions (and also from the production order enquiry). You can select the sales order line and see the production order enquiry. This facility is also available in reverse from the production order enquiry.

On booking in the finished product, the system automatically allocates the quantity to the associated sales order line, allowing it to be subsequently picked and despatched (using the batch, lot or serial number as its unique reference).

Enquiries

There are three particular types of enquiry which are relevant to a configuration environment, i.e.:

- The status of a particular order
- The traceability of a particular configuration throughout the processing cycle
- Details of what is contained within a particular configured product

Reports

All of the standard sales and production reports and documents are available for configured products.

The lot/batch/serial number, if used, are printed and provide access to the particular product configuration. Details of the configured options are included on pick lists, invoices, etc.

In addition, listings are available (with selection criteria) for the following:-

- Rules
- Messages
- Items by search family characteristics

Restrictions

- Although the standard Production modules are available in two modes - Base and Extended - the Configurator uses only Base mode.
- Whether your system is normally Base or Extended, as soon as you begin to use Configurator, all routes created with the configuration route automatically use Base mode and functionality.
- Configured items cannot be either kit components (because they need to be priced in Order Entry) or kit parents (because stock needs to be held against them during order processing).
- The Sales Analysis application does not distinguish between sales of different configurations for the same item. For this reason, Configurator maintains an analysis file, CNP90, which holds details of the different configurations. You can use Query or Viewfinder or another file manipulation function to produce an analysis.



Warning: You should take care when setting up rules. Complex rules systems are difficult to implement and any amendment is liable to lead to unforeseen problems.

Implementation Plan

This section is a guide to the planning and implementation of Configurator, and lists the prerequisites of the module, outlining the recommended implementation procedure.

The various stages need to be thought through and an implementation plan produced. The configuration designers need to be thoroughly familiar with the routes and assemblies to be configured, and you must spend time planning items which are to be generic, which options are to be taken at preconfiguration and which at order configuration, and so on. The implementation plan should allow for the training of users, configuration planning, and data entry.

The list below shows the modules involved:

Process	Initiating Module
Set up the item data	(Several)
Create rules	Configurator
Create configured items	Configurator
Select product on customer order line	Advanced Order Entry
Select mandatory components	Configurator
Select optional operations/components	Configurator
Create customer order line	Advanced Order Entry
Create/amend production order	Advanced Order Entry
Release production order	Production Control
Issue material	Production Control
Report operations	Production Control
Receive finished goods	Production Control
Allocate to customer order	Production Control

Implementation Procedure

After installing the software, the recommended steps for implementing Configurator are as outlined below. Each of the steps is detailed within other sections, the list given here is intended to be used in conjunction with the detailed sections on each option.



Note: The option Configurator Application Manager Installation on the Advanced Order Entry Utilities menu (1/AOU), must be run after installing Configurator. The option only needs to be run once. This is to set up library lists, tasks, menus etc. This is because the configurator objects exist in the Advanced Order Entry libraries.



Note: Company details and reference data (work stations, stockrooms etc.) must be set up in all prerequisite applications before defining Configurator details.

A note on menus: some Inventory and Production options appear on Configurator menus for your convenience. The Inventory item master option, for example, can be taken from within Inventory, Configurator or Production. It makes no difference which you use.

The implementation guide instructs you as to when to set up Inventory Description file entries, but for general information the complete list is as follows:

- Configurator rule categories (RCAT)
- Configurator message categories (MCAT)
- Inventory Sources (INSC)
- Operation codes (OPCD)
- Ordering Methods (ORDM)
- Characteristics (user-defined)
- Configurator operation IDs (CNOI)

Implementation Steps



This section takes you through the steps necessary to set up Configurator once you have installed it. You will find it useful to read these instructions and carry them out in order.

Each numbered heading gives the menu option you must enter, then gives details of what needs setting up.

Read the instructions carefully, and ensure you understand the terms used. If you are unsure, reread the relevant section of this overview, or see the InfoBase Configurator glossary.

1. The Descriptions option in Inventory Management



1/INM, OR 21/CNM

Set up a description table for Inventory Source (INSC), defining a specific and unique value for each of the following item categories in Configurator:

- Model items
- Base items
- Preconfigured items
- Order configured items
- Generic components



The codes M, B, P, O and G are suggested simply because they are the initial letters of each category, and so will be easy to remember, but if you already use these INSC codes for another purpose then you must choose different ones.

2. The Descriptions option in Inventory Management



1/INM, OR 21/CNM

If selection by characteristic is to be used, set up a new major type (within the Inventory descriptions file) for each characteristic, and add there all the associated values for each characteristic required. If the characteristic was colour, for example, you might set up major type COLR with blue, red and green as values within it.

A new, separate major type must be set up for characteristic you plan to use.



When entering the codes, you must type V in the 'Percentage/Value' field for the major type, otherwise the codes are not picked up for Configurator prompts.

3. The Descriptions option in Inventory Management



1/INM, OR 21/CNM

Set up an Inventory descriptions table for ordering methods (ORDM). You must set up the following specific values:

- 0 - Normal stocked item.
- 1 - Make-to-order.
- 2 - Purchase-to-order.
- 3 - Deliver direct.
- 4 - Reserve.
- 5 - Configure-to-order.

4. The Descriptions option in Inventory Management

1... 1/INM, OR 21/CNM
2...

Set up an Inventory descriptions table for rule categories (RCAT) and message categories (MCAT).

Every rule and message you use must be assigned to a category. One RCAT category and one MCAT category is enough unless you wish to use category as a search facility. This would be useful if you use rules extensively.

Note that the codes you set up must not be more than 5 characters long.

5. The Descriptions option in Inventory Management

1... 1/INM, OR 21/CNM
2...

Set up Inventory descriptions for the Configurator operation ID (CNOI).

A unique operation ID must be assigned to each operation within a Configurator route. This must not be unique simply within its own route, but unique for the company. The code is entered when you define the route, and is mandatory.

6. The Search Family/Characteristics option in Configurator

1... 3/CNM
2...

You can only begin this next stage when you have set up a major type in Inventory for each characteristic you are going to use (see above).

This step involves setting up the search family codes you are using, defining up to ten characteristics for each. Search family groups are used by Configurator for several purposes:-

- To link a model product with associated base products.
- To link a generic component with its actual components.
- To search for a matching component based on the specified characteristics.

Thus each model has its own search family, whose code appears in the Inventory item master of each of its related base products.

The same is true of each generic, which is linked to its actuals.

7. The Company Profile Maintenance option in Configurator

1... 1/CNU
2...

Define the Configurator company profile. The critical settings here include:

- The production route code for configurations
- Item categories (Inventory source codes) for model, base, preconfigured and order configured items, and generic components.
- The pricing method to be used for the configured item.

8. Either the Items option or the Linked Item Maintenance option, both in Inventory Management

1...
2...

2/INM, 10/INM

Set up Inventory item master records for model items and base items, assigning the 'Inventory Source' flag as defined for each type of item on the Configurator company profile.



The Linked Item Maintenance option enables you to enter all item maintenance details via a single menu option, instead of having to switch between menus and options.

Inventory offers a 'base on' facility (Copy Based On Item Creation from the Inventory Maintenance menu) which creates a new item based on an existing one, copying not only the Production details but Inventory item master and stockroom details, as well as other details as requested.

Each item can be set up with a default ordering method. This defines the usual method of sourcing for the item when ordered in Advanced Order Entry.

Group together related model and base items by assigning them the same search family code. Optionally define appropriate item search characteristics for each base item.

9. Either the Items option or the Linked Item Maintenance option, both in Inventory Management

1...
2...

2/INM, 10/INM

Set up Inventory item master records for all generic and actual components, assigning the 'Inventory Source' flag for generic components only.

Each item may be set up with a default ordering method. This defines the usual method of sourcing for the item when ordered in Advanced Order Entry.

Group together related generic and actual components by assigning them the same search family code. Define appropriate item search characteristics for each actual component by using the **F18** key.

10. The Stockroom Details option or the Linked Item Maintenance option, both in Inventory Management

1...
2...

5/INM, 10/INM

Set up Inventory item/stockroom records for every model and base item, and every generic and actual component.

Configured items sold within Advanced Order Entry need to be defined to the sales depot specified for the configured order line.

11. The Production Details option in Production Definition Management, or the Linked Item Maintenance option in Inventory Management

3/MDM, 10/INM

Set up Production item master records for every model and base item, and every generic and actual component.

Phantoms are supported by Configurator.

12. The Configurator Rules option and the Configurator Messages option, both in Configurator

1/CNM, 2/CNM

Set up component and operation rules in Configurator, with corresponding messages if required. This requires Description file codes to be set up for rule categories (RCAT) and message categories (MCAT). No special values are required for these codes.

A set of rules may be defined within the application to check the validity of combinations of operations, components and text formats for particular configurations. The rules operate 'intelligently' during configuration, performing automatic inclusions and exclusions as the configuration is built.

Specific messages may be defined within the application, to be applied either to particular Configurator rules or to maximum and minimum component quantities for a configuration.

When a particular rule is broken, or the 'quantity per' entered for an component is outside the permitted range, the appropriate Configurator message is displayed. The rules and messages may be split into categories which aid their search and selection, particularly if there are a large number defined to the Configurator.

13. The Text Types option in Production Definition Management

41/MDM

If Configurator formatted operation text is required, set up text formats in Production Text Types.

14. The Routes/Structures option in Production Definition Management

21/MDM

Set up a production route for the model item or order configured item, using the configuration route code as defined on the Configurator company profile. Define all possible components and operations for the model item. Define whether each one is optional or mandatory and whether it is selectable during preconfiguration or at order entry time. Optionally define text formats and Configurator rules to the route.

15. The Preconfiguration option in Configurator

1/CNP

If you are using the preconfiguration option:

Generate preconfigured items in preconfiguration from model/base item combinations.

16. The Price/Discount Profiles option, the Price Lists option, and the Discount Lists option, all in SOP

1... 4/OEM, 5/OEM, 6/OEM 2...

Set up your pricing details. This depends on the pricing method you have selected in the Configurator company profile:

Pricing method - 0:

Set up the price list and discount list for the parent item only.

Pricing method - 1:

Set up the price and discount lists for all components.

Pricing method - 2:

This method requires several setup steps:

- a) Define operation costs on the Production route.
- b) Run the option (2/MDC) to write Production costs to a file accessed by Configurator.
- Both a) and b) are used only if the company profile specifies the use of operation costs.
- c) Create a cost price list profile for components.
- d) Run the option in Configurator to write costs to the cost price lists.
- e) Set up the cost discount profile for components, if discounts are to be applied.
- f) Set up the mark-up percentage for the parent item. This is done in the cost discount list, using a negative value.

See Maintenance for further details.

17. The Configuration options in Configurator

1... 2/CNP, 3/CNP 2...

Take orders for configured items in Advanced Order Entry, either by directly entering the configured item code, or by entering a base item with search type 9 to select from a list of all configured items related to the base item. This allows the configuration to be set by selection of the remaining options. When the sales order is updated, corresponding production orders are generated to manufacture configured lines.

18. Standard Production Control options.

Process the production orders through Production Control as for normal production orders. When the completed configured items are booked into Inventory, stock is automatically allocated to the configured sales order lines.